

Grade 6 Accelerated  
Unit 14  
Lesson 1 Homework

Name \_\_\_\_\_  
Date \_\_\_\_\_  
Period \_\_\_\_\_

According to Florida Building Code, a parking space must be no less than 12 feet wide.

1. Write an inequality to represent the width,  $w$ , that parking spaces can be.
  
  
  
  
  
  
  
  
  
  
2. Represent the inequality on a number line.
  
  
  
  
  
  
  
  
  
  
3. List two examples of possible widths, in feet, of the parking spaces.
  
  
  
  
  
  
  
  
  
  
4. Give one example of a width that is true for the inequality but does not make sense in the real world.

The maximum weight for a rider on a collapsible-floor water slide is 400 pounds.

5. If  $w$  represents the weight of the rider, write an inequality to describe all possible values of  $w$ .

6. Represent the inequality on a number line.
7. Does the inequality that represents the situation include 0 as a possible weight? Why or why not?
8. Explain why a negative number does not make sense in this context.

A farmer plans to plant more than 56.75 acres of lemons.

9. If  $L$  represents the number of acres of lemons the farmer plants, write an inequality that represents the number of acres the farmer plants.
10. Represent the inequality on a number line.

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Unit 14  
Lesson 2 Homework

Name \_\_\_\_\_  
Date \_\_\_\_\_  
Period \_\_\_\_\_

Solve.

1.  $x - 14.5 = 41.7$

2.  $\frac{k}{6} = 12.25$

3.  $0.75 - p = 0.19$

4.  $18 = 0.5t$

5.  $6.5 + f = 17.11$

6.  $\frac{2}{3}r = 22.8$

7. Mike made jelly and filled three equal sized jars with  $3\frac{2}{3}$  cups of jelly. How much jelly did Mike make?
8. Davis cut  $\frac{1}{6}$  feet off of a piece of fabric, leaving  $4\frac{1}{2}$  feet of fabric left. How long was the fabric before Davis cut it?
9. An environmental science teacher has a 10-gallon fish tank. Currently, the tank has  $4\frac{9}{11}$  gallons of water inside. How much more water is needed to fill the tank?
10. The 6<sup>th</sup> grade class at a school is volunteering to decorate a hallway for homecoming. The students have decorated  $\frac{5}{16}$  of a mile in decorations which is  $\frac{3}{4}$  of the hallway. What is the length of the hallway being decorated?

Grade 6 Accelerated  
Unit 14  
Lesson 3 Homework

Name \_\_\_\_\_  
Date \_\_\_\_\_  
Period \_\_\_\_\_

1. What is the inverse of addition?
2. What is the inverse of subtraction?

For questions 3-7, use the algebra tiles shown.

The diagram shows an equation represented by algebra tiles. On the left side of the equation, there are five small squares, each containing the number -1, arranged in two rows: the top row has five tiles and the bottom row has four tiles. In the middle is an equals sign. On the right side, there are seven small squares: four in the top row (each containing -1) and three in the bottom row (each containing -1). To the right of these tiles is a tall vertical rectangle containing the letter  $x$ .

3. What expression represents the left side of the equation?
4. What expression represents the right side of the equation?
5. Write the algebraic representation of the equation.
6. What operation would be used to solve the equation?

7.  $y + 5 = 17$

8.  $-8 + x = 66$

9.  $-6 + x = -10$

10.  $14 + x = 18$

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Lesson 4 Homework

Name \_\_\_\_\_  
Date \_\_\_\_\_  
Period \_\_\_\_\_

Solve.

1.  $17 = 10 + x$

2.  $x - 17 = 231$

3.  $-229 = -20 + h$

4.  $313 = g - 91$

5.  $y - 6 = -19$

6.  $18 + u = -3$

7.  $y + 9 = 31$

8.  $8 + x = -77$

9.  $-13 + x = 117$

10.  $41 + x = 809$

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Lesson 5 Homework

Name \_\_\_\_\_  
Date \_\_\_\_\_  
Period \_\_\_\_\_

Write and solve an algebraic equation for each of the following.

1. Hiro has 18 highlighters, which is 4 fewer than Martin has. How many markers does Martin have?
  
  
  
  
  
  
  
  
  
  
2. A college engineering program receives 1,245 applications this year, of which only 200 can be accepted. How many students are not accepted?
  
  
  
  
  
  
  
  
  
  
3. Pepper has an 8 pack of markers. She let Shawn borrow some of her markers from the box. When she counted the markers in the box again, she had 5 remaining. How many markers did Shawn borrow?
  
  
  
  
  
  
  
  
  
  
4. Mrs. Tuttle took her class to the media center for their class photo. Of her 18-student class, only 13 were present for picture day. How many students were absent?
  
  
  
  
  
  
  
  
  
  
5. Jasmine consumed a total of 1,450 calories of her 2,000 daily calorie limit. How many more calories can Jasmine consume?

6. A grocery store set out 78 apples on a display stand. At the end of the day, there were 31 apples remaining. How many apples were sold?
7. A soccer team is doing a car wash to raise money. The team earned 395.45 between cars and donations. If \$176.40 was raised from washing cars, how much was made from donations?
8. Kiegan collected 9 more items than his cousin Joel did for the annual toy drive. Joel collected 15 items. How many items did Kiegan collect for the toy drive?
9. A custom bench was delivered and turned out to be too long. 4 inches were removed from the bench to make the new length 36 inches in length. What was the original length of the bench, in inches, when it was delivered?
10. Sarah was sharing a story about her favorite birthday party from 8 years ago. If Sarah is now 21, how old was she at her favorite birthday party?

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Unit 14  
Lesson 6 Homework

Name \_\_\_\_\_  
Date \_\_\_\_\_  
Period \_\_\_\_\_

1. What is the inverse of multiplication?
2. What is the inverse of division?

For questions 3-7, use the algebra tiles shown.



The diagram shows an equation represented by algebra tiles. On the left side, there is one dark gray tile labeled  $\frac{x}{6}$  followed by five light gray tiles, for a total of six tiles. An equals sign is in the middle. On the right side, there are three dark gray tiles, each labeled with the number 1, for a total of three tiles.

3. What expression represents the left side of the equation?
4. What expression represents the right side of the equation?
5. Write the algebraic representation of the equation.
6. What operation would be used to solve the equation?
7. What value of  $x$  makes the equation true?

8.  $-4 = \frac{f}{11}$

9.  $-6 = \frac{y}{2}$

10.  $-14x = -56$

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Lesson 7 Homework

Name \_\_\_\_\_  
Date \_\_\_\_\_  
Period \_\_\_\_\_

Solve.

1.  $25 = \frac{j}{-5}$

2.  $6x = 42$

3.  $\frac{h}{5} = 6$

4.  $-440 = 44m$

5.  $-34 = 2x$

6.  $21 = -\frac{y}{9}$

7.  $\frac{r}{4} = 72$

8.  $-12p = 36$

9.  $-3 = \frac{y}{5}$

10.  $-x = 56$

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Lesson 8 Homework

Name \_\_\_\_\_  
Date \_\_\_\_\_  
Period \_\_\_\_\_

Write and solve an algebraic equation for each of the following.

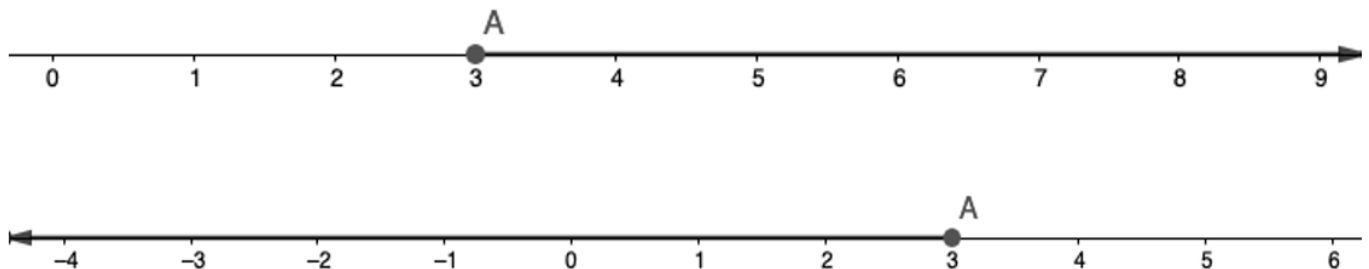
1. At a restaurant, 6 friends decided to split the bill evenly. If each person paid \$14, what was the total bill?
  
  
  
  
  
  
  
  
  
  
2. A Great White Shark has 300 teeth. If the teeth are evenly arranged into 3 rows of teeth, how many teeth are in each row?
  
  
  
  
  
  
  
  
  
  
3. A school has a budget of \$400,000 for support staff members. If each support staff member makes \$40,000 each, how many support staff members can the school employ?
  
  
  
  
  
  
  
  
  
  
4. During a math adoption, there were 4 textbook company finalists invited to present their materials to the school district. For spacing purposes, each company could only bring a maximum of 3 people each. What is the maximum amount of people that could attend from the textbook companies?
  
  
  
  
  
  
  
  
  
  
5. A princess dress takes 6 yards of fabric to create. If an order is placed for 9 dresses, how many yards of fabric will be needed?

6. A certain apple tree can produce 210 apples at a given time. If the apples are placed into bins that can hold 21 apples each, how many crates will be needed for the entire tree?
7. A recipe for Red Velvet cake calls for 1.5 sticks of butter for each cake. If someone wants to make 4 cakes, how many sticks of butter will be needed?
8. A new home builder offers 6 different floorplans, each that come with 5 color pallet options per floorplan. What is the number of different options that a buyer has with this builder?
9. A garment moving box can hold 8 suit jackets. If 4 garment boxes can fit in the back of a vehicle, what is the maximum number of suit jackets that can be moved in one trip?
10. Thelma is participating in a Walk-a-Thon where she will get \$0.25 per lap that she walks. How many laps would she need to walk to reach a goal of \$50?

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Unit 14  
Lesson 9 Homework

Name \_\_\_\_\_  
Date \_\_\_\_\_  
Period \_\_\_\_\_

1. For each inequality, write three numbers that could represent the solution set.
  - A.  $x > 9$
  - B.  $-12 \leq m$
  - C.  $g < 0$
2. The Overseas Highway, also known as the "Highway that Goes to Sea," runs 113 miles from Miami down to Key West. The road connects the Florida Keys to the Florida mainland. The speed limit of the road is 55 miles per hour. Represent the speeds that one could drive without exceeding the speed limit in an inequality with  $s$  representing the speed.
3. Graph the solution  $x \geq 12$ .
4. Chris said that the numbers  $(-9, -10, -11)$  be in the solution set for the inequality  $r < -9$ . Explain if Chris' answer is correct.
5. Compare and contrast the two number lines.



6. Use the inequality  $-9.75 > y - 2.5$  to complete the following:

A. Find the solution set by isolating the variable. Show your work.

B. Graph the solution set on the number line.

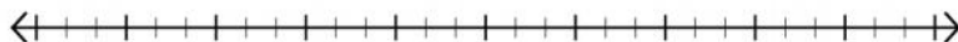


C. Use substitution to test a value or values in order to verify that the solution and the graph are correct.

7. How do you isolate a variable when solving an inequality?

Solve each inequality. Then, graph its solution set.

8.  $-15\frac{1}{4} \leq m + 6.5$



9.  $48 > y - 24$



10.  $21.4 + w < 56.9$



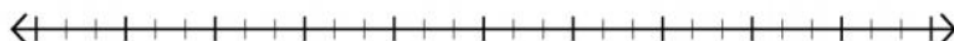
Grade 6 Accelerated  
Unit 14  
Lesson 10 Homework

Name \_\_\_\_\_  
Date \_\_\_\_\_  
Period \_\_\_\_\_

- When flying on an airplane, your travel bag can weigh no more than 50 lbs. Your bag currently weighs 27.3 lbs. The inequality  $27.3 + m \leq 50$  represents the situation, where  $m$  represents the amount of weight that can be added to your bag. What is the maximum amount of weight that can be added to your bag while still making the weight requirement?
- You earn \$8.75 per hour for walking dogs in your neighborhood. You are saving up your dog walking money to purchase a new kayak that costs \$280. The inequality  $\$8.75h \geq \$280$  represents the situation, where  $h$  represents the number of hours worked. What is the minimum number of hours you must walk in order to purchase the kayak?
- Complete the table by substituting each value of  $x$  to determine if that is a solution to each inequality.

| Value of $x$   | What is the value of $-4.5x$ | Is $-4.5x > 18$ true or false? | Solution?                                             |
|----------------|------------------------------|--------------------------------|-------------------------------------------------------|
| -4             |                              |                                | <input type="radio"/> Yes<br><input type="radio"/> No |
| -1.5           |                              |                                | <input type="radio"/> Yes<br><input type="radio"/> No |
| 3              |                              |                                | <input type="radio"/> Yes<br><input type="radio"/> No |
| $6\frac{1}{5}$ |                              |                                | <input type="radio"/> Yes<br><input type="radio"/> No |

- Graph the inequality on the number line.



5. Find the solution to both inequalities using the division property of inequality.

|                |               |
|----------------|---------------|
| $-7y \leq -35$ | $7y \leq -35$ |
|----------------|---------------|

6. Find the solution to both inequalities using the multiplication property of inequality.

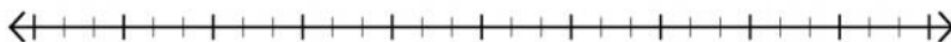
|                   |                    |
|-------------------|--------------------|
| $\frac{y}{3} > 2$ | $\frac{y}{-3} > 2$ |
|-------------------|--------------------|

Solve each inequality. Then, graph its solution set.

7.  $\frac{r}{-4} < 7$



8.  $-0.4t \geq -12$



9.  $\frac{m}{-2} > \frac{5}{2}$



10. Explain why the direction of the inequality symbol must be reversed when multiplying or dividing by the same negative number.